

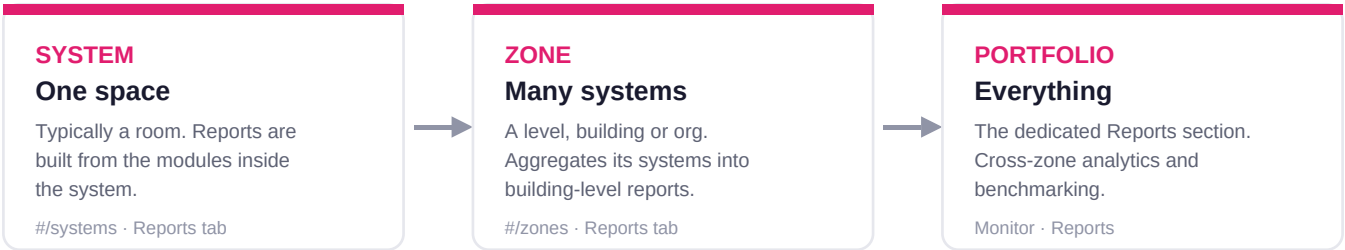
Reports: one analytics layer, three levels of zoom

Reports gives operators and workplace teams answers about how spaces are actually used — without leaving Backoffice. The same reporting layer renders at three scopes that mirror the PlaceOS hierarchy: a single **system**, a **zone** of systems, and the whole **portfolio**. This brief describes the model, what each level is for, and one open proposal we'd like the product team's view on.

Engineering · July 2026 · Prototype running in Backoffice with sample data

The model: scope follows the hierarchy

PlaceOS already organises the world into systems (spaces) grouped by zones (levels, buildings, organisations). Reports doesn't invent a new mental model — it attaches an analytics view at each level of that existing one. The closer to a single space you are, the more operational the report; the further out you zoom, the more strategic it becomes.



	SYSTEM REPORTS	ZONE REPORTS	DEDICATED SECTION
Scope	One system — typically a room, but also plant, signage or any control space	All systems in a zone — a level, a building, an org	Any zone, portfolio-wide
Question it answers	“How is this room performing?”	“How is this building performing?”	“How do we compare, and where should we invest?”
Content driven by	The modules inside the system	Aggregation across member systems	Cross-zone analytics, benchmarking, AI insights
Audience	Operators, support, AV/FM techs	Building & workplace managers	Workplace leads, executives

System level: module-aware reports for a single space

The Reports tab on a system is about **that system only**. Most of the time that means a room — a meeting space, an activity space — but the same applies to any system: a lecture theatre, a plant room, a signage network.

The important idea is that the tab is **module-aware**. A system's reporting isn't configured; it's *derived*. We inspect which logic modules the system runs and compose the view from template report blocks, one per capability:

LOGIC MODULE PRESENT	TEMPLATE REPORT VIEW
Location services / people counting	People count & occupancy: live count vs capacity, peaks, daily patterns, dwell
Bookings / meeting logic	Booking data: meetings today, utilisation vs bookings, no-shows & ghost meetings, peak hours
Environment / sensors	Climate & air quality: temperature, humidity, CO ₂ , comfort score
Power / energy logic	Consumption: current load, daily usage, lighting & HVAC state
(future) Any driver that opts in	Drivers declare a report template in metadata; the tab renders it automatically

Zero configuration is the feature

Add a people-counting module to a room and an occupancy report appears on that room. Remove the booking module and the booking report goes away. The reporting surface always tells the truth about what the system can measure — and it gives our drivers team a clear contract: ship a logic module, get a report template with it.

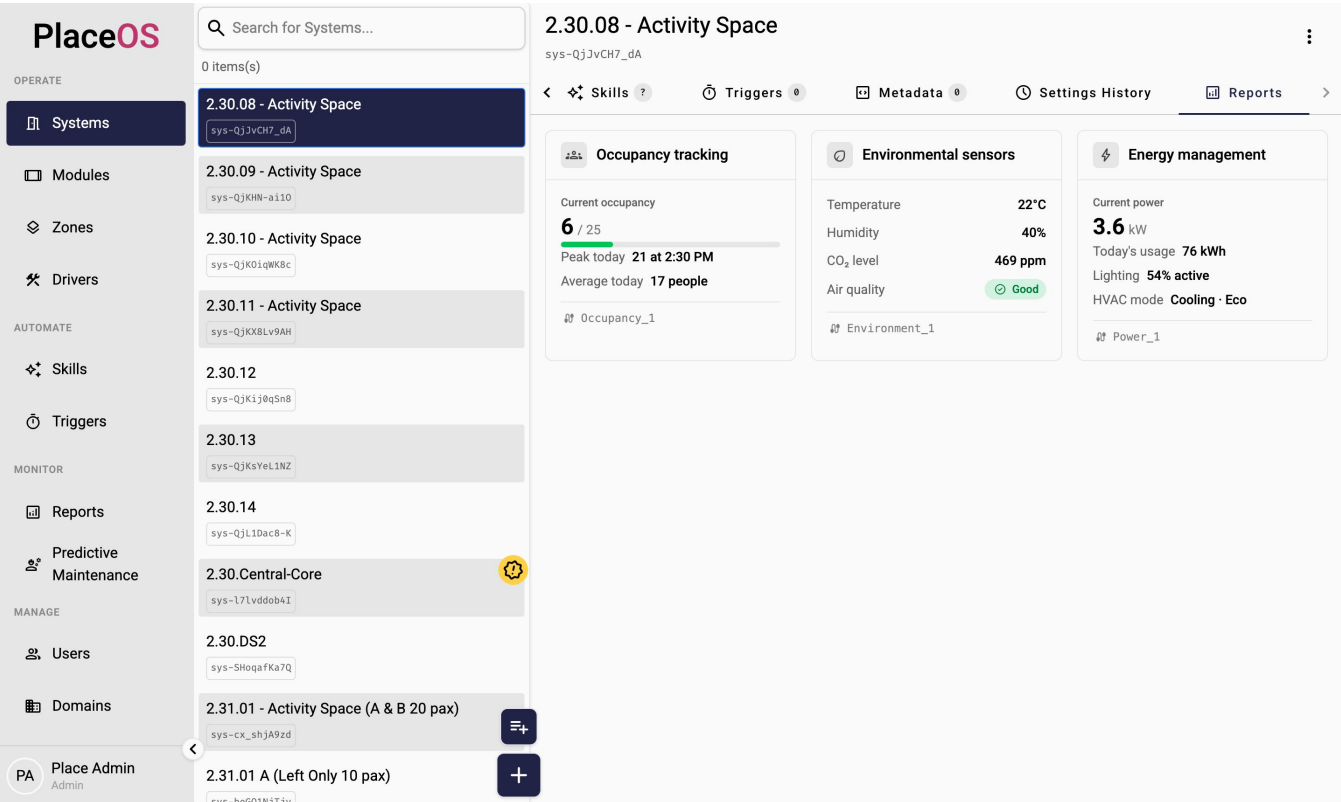


Fig 1 — System Reports tab (prototype). This system runs occupancy, environment and power modules, so it gets those three template views. Each card footers the module it reads from (e.g. Occupancy_1) — the same binding convention used by Skills.

Zone level: a building view assembled from its systems

The Reports tab on a zone summarises **many systems at once**. Where the system tab is operational (“is this room too warm?”), the zone tab is managerial (“how utilised is Level 30 this week?”). Because zones nest — level → building → organisation — the same tab scales from a floor summary to a whole-building rollup.

Zone reports lead with a KPI strip (systems, live occupancy, active meetings, energy today) and expand into building-flavoured report families:

- **Occupancy trends** — peaks, weekly averages, busiest days across the zone
- **Meeting analytics** — volume, duration, no-shows, most popular rooms
- **Environmental health** — temperature, air quality and comfort across spaces
- **Energy performance** — load vs target, efficiency, savings opportunities
- **Space utilisation** — desks vs meeting rooms vs common areas
- **System health** — online counts, alerts, incidents (operations at a glance)

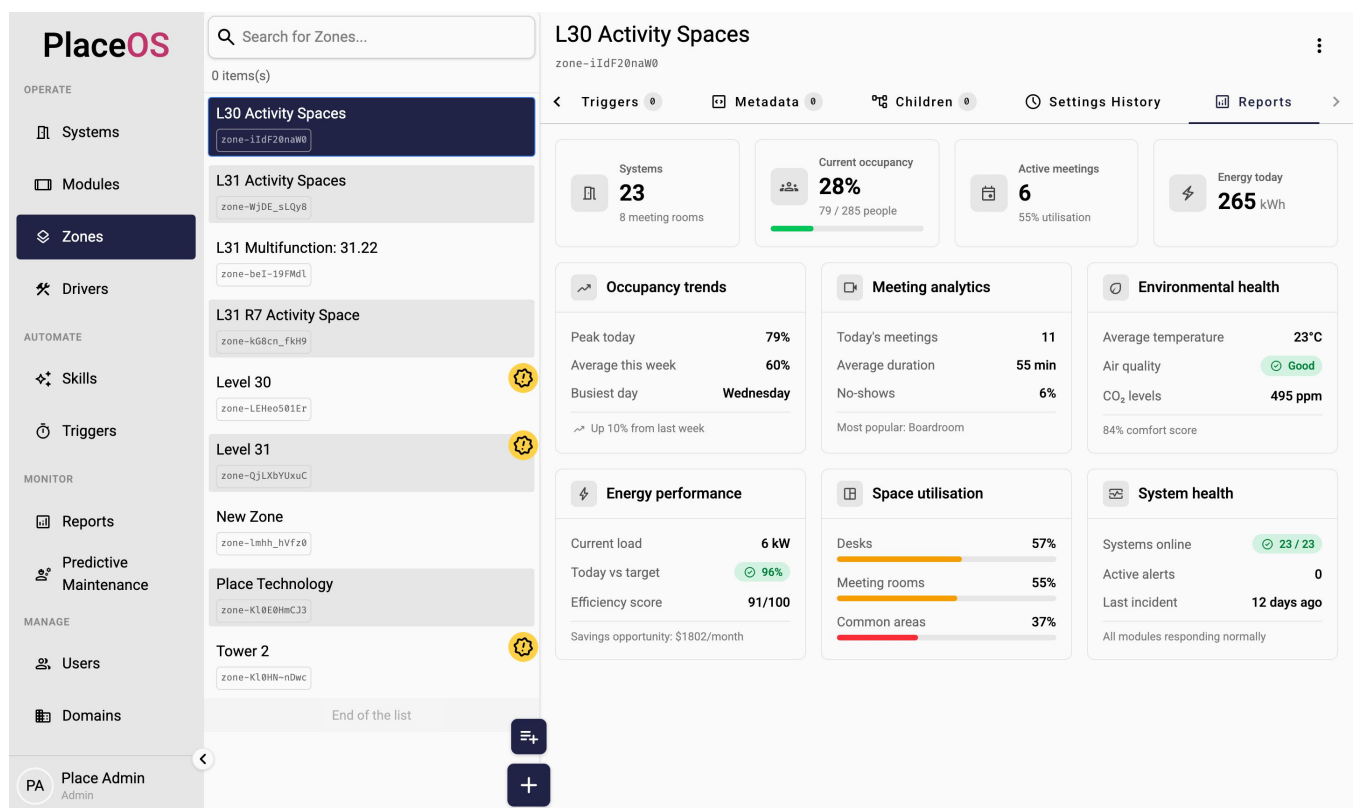


Fig 2 — Zone Reports tab (prototype) on a level: KPI strip plus six report families aggregated across the zone's 23 systems.

The same module-awareness applies in aggregate: a zone whose systems have no power logic shouldn't show an energy card. Zone reports are the sum of what their systems can measure.

The dedicated Reports section: the portfolio lens

Reports also lives in the main navigation (under **Monitor**) for people whose job starts with the reports rather than with a particular room. You pick a zone — organisation, building or level — and the section **defaults to that zone's reports**: the same building view as the zone tab, reached from the other direction. From there it adds what only a portfolio view can do:

- **AI Analytics** — occupancy prediction, energy anomaly detection, meeting intelligence and space-optimisation recommendations for the selected zone
- **Benchmarking** — buildings scored and ranked against each other, KPIs vs targets, and industry benchmarks (energy intensity, water, waste, air quality)

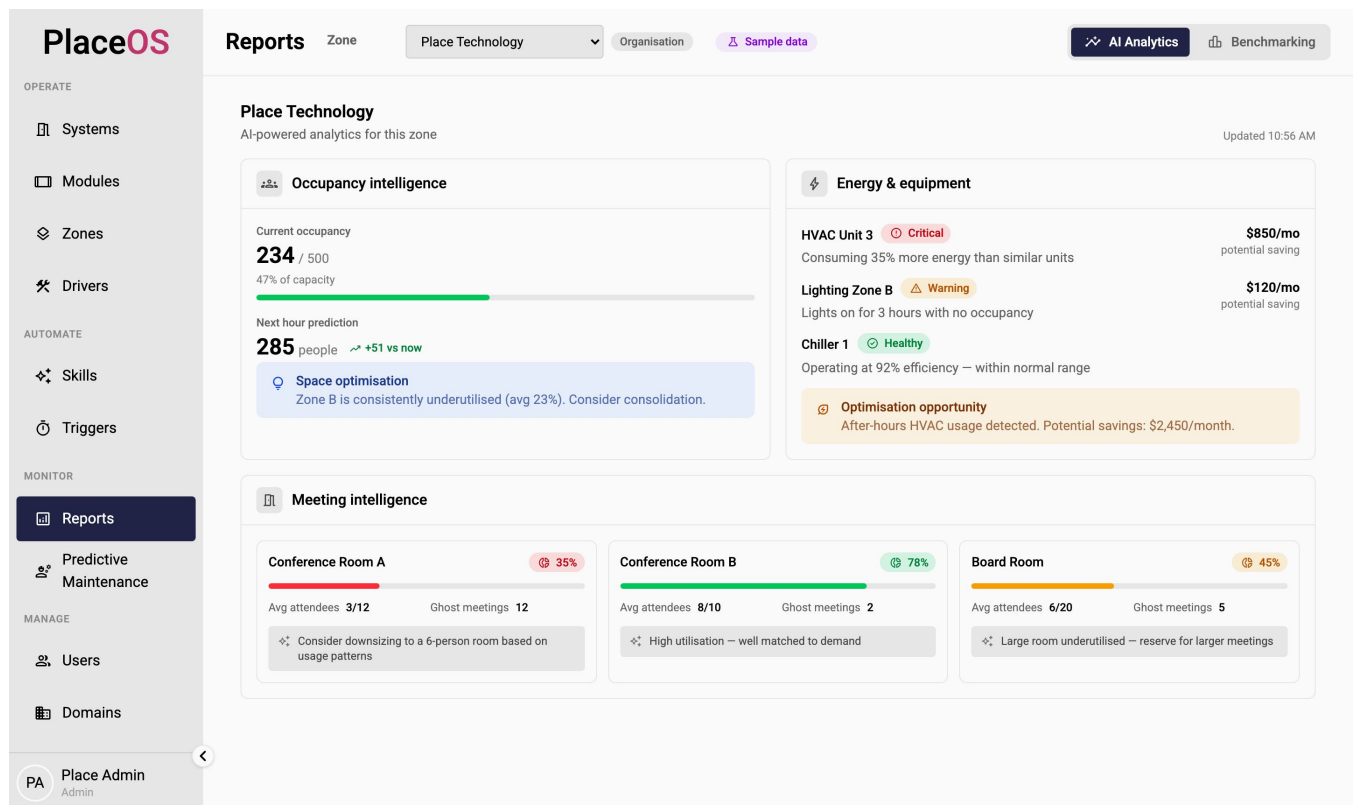


Fig 3 — Dedicated Reports section (prototype): zone selector, and the AI Analytics / Benchmarking views for the selected zone.

For discussion: saved reports as the section's library

There's a natural question the three-level model raises: what is the dedicated section *for*, once zone and system tabs exist? Our proposal — and the thing we'd most like product input on — is that it becomes the **home for saved reports**.

The flow we have in mind:

- You're on a zone or system Reports tab, and you've shaped a view worth keeping — a date range, a set of cards, a comparison.
- You hit **Save report**. It gets a name, and lands in the dedicated Reports section.
- The section becomes a library: your saved reports, pinned portfolio dashboards, and (later) scheduled exports — rather than only a live explorer.

Why this might be right

It mirrors how people actually work: reports are discovered in context (“this room, this week”) but consumed on a cadence (“every Monday, for my exec”). Saving bridges the two. It also gives the dedicated section a clear identity — the place where reporting *artefacts* live — instead of duplicating the zone tab.

Open questions for the team

Are saved reports personal, shared per-team, or org-visible? Do they snapshot data or stay live? Do we fold in scheduling/PDF export at the same time or later? And does “Save report” belong on every card, or on the tab as a whole?

Where the prototype is today

- All three surfaces are **built and navigable** in Backoffice: system tab, zone tab, and the dedicated section (AI Analytics + Benchmarking), sharing one widget system so they look and behave identically.
- Module detection currently keys off system contents with simple heuristics; the target is drivers/logic modules **declaring their report templates** so detection is exact.
- All report **data is generated sample data**, clearly flagged in the UI. Wiring real sources (module state, booking APIs, metrics store) is the next engineering step and doesn't change any layout.
- Predictive maintenance was split out to its own Monitor section — reports stay descriptive; actionable alerting lives there.

None of the ideas in this document are load-bearing yet — the point of the prototype is to make this conversation concrete. Feedback welcome on the saved-reports proposal in particular.